

EX 12:

Implement Iris Flower Classification using KNN

PROGRAM :

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
# KNN
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Accuracy:",
      accuracy_score(y_test, y_pred))
```

```
# New flower
sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("Predicted Flower:",
      iris.target_names[prediction[0]])
```

OUTPUT:

```
Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Accuracy:
1.0

New Flower Prediction:
setosa
```